FINC5001

Workshop 6

Capital Asset Pricing Model (CAPM)

Presented by
Craig Mellare
University of Sydney Business School

THE UNIVERSITY OF SYDNEY
Business School

Week 7 Workshop Quiz (Risk-adjusted Discounted Cash Flows, Investor Preferences, Diversification and Portfolios)

Time – 10 minutes

THE UNIVERSITY OF SYDNEY
Business School

Module 6 – Capital Asset Pricing Model (CAPM)

Brief Recap

THE UNIVERSITY OF SYDNEY
Business School

Beta

- The formula for beta is:

$$\beta_i = \frac{Cov(R_m, R_i)}{\sigma_m^2}$$

- β_i = Beta for stock i
- R_m = Returns for the market
- R_i = Returns for stock i
- σ_m^2 = Variance for the market

- Stocks with beta > 1 tend to amplify the overall movements of the market
- Stocks with beta < 1 tend to dampen the overall movements of the market
- Stocks with beta = 1 tend to move 1-1 with the market

Beta

- **Portfolio beta** measures the systematic risk of a portfolio.

- Portfolio Beta (2 stocks)

$$\beta_p = \omega_1\beta_1 + \omega_2\beta_2$$

- β_p = Portfolio beta
- ω_1 = Weight on stock 1
- ω_2 = Weight on stock 2
- β_1 = Beta of stock 1
- β_2 = Beta of stock 2

Beta

- Portfolio Beta (n stocks)

$$\beta_p = \omega_1\beta_1 + \omega_2\beta_2 + \cdots + \omega_n\beta_n$$

- β_p = Portfolio beta
- ω_i = Weight on stock i
- β_i = Beta of stock i

- **The portfolio beta is the weighted average of all the individual betas.**

CAPM

- The capital market line (CML) turns out to lead to the CAPM.
- The CAPM equation is:

$$ER_i = r_f + \beta_i \left(ER_m - r_f\right)$$

$$\beta_i = \frac{Cov(R_m, R_i)}{\sigma_m^2}$$

- ER_i = Expected return of stock i
- r_f = Risk-free rate
- β_i = Beta for stock i
- ER_m = Expected return of market

CAPM

– What does this mean for the expected returns for a stock?

Beta = 1 means $ER_i = ER_m$

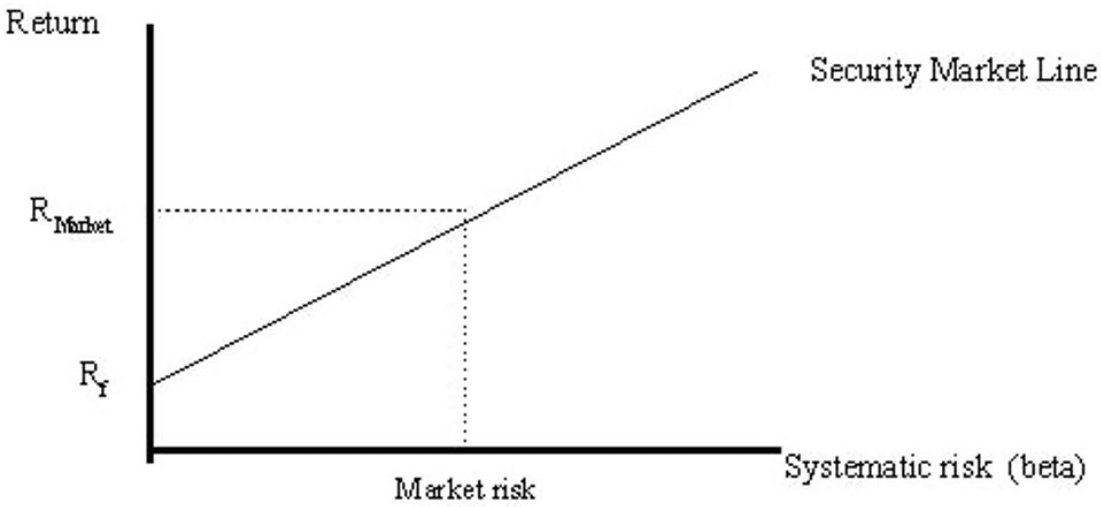
Beta = 0 means $ER_i = r_f$

Beta > 1 means $ER_i > ER_m$

Beta < 1 means $ER_i < ER_m$

– We graph this relationship:

$$ER_i = r_f + \beta_i\left(ER_m - r_f\right)$$



The line is called the **security market line (SML)**.

Performance Evaluation

	Definition	Application
Sharpe ratio	$\frac{Excess\ return}{Standard\ deviation}$	Choosing among portfolios competing for the overall risky portfolio
Treynor ratio	$\frac{Excess\ return}{Beta}$	Ranking many portfolios that will be combined together to form the overall risky portfolio
Jensen's alpha	$Actual\ average\ return - return\ predicted\ by\ CAPM$	Measuring performance compared to the market portfolio

APT

For 2 factors, the APT model is:

$$ER = r_f + \beta_1(r_1) + \beta_2(r_2)$$

ER = Expected return of the stock

r_f = Risk-free rate

β_1 = Beta for factor 1

β_2 = Beta for factor 2

r_1 = Risk premium for factor 1

r_2 = Risk premium for factor 2

APT

For n-factors, the APT model is:

$$ER = r_f + \beta_1(r_1) + \beta_2(r_2) + \cdots + \beta_n(r_n)$$

ER = Expected return of the stock

r_f = Risk-free rate

β_i = Beta for factor i

r_i = Risk premium for factor i

Workshop Questions

THE UNIVERSITY OF SYDNEY
Business School

Question 1

Hansel and Gretel escaped with all the witch's wealth and invested 70% of it in Adidas (ADS) and the remainder in Volkswagen (VOW) on the German Stock Exchange (Deutsche Borse Xetra).

- They used the daily returns of the most recent six months to calculate the risk (SD) of each share at 1.794% and 1.251%, respectively.
- From a relevant market portfolio, the expected daily return is 0.038% and the standard deviation of returns is 1.006%.
- The risk-free rate is 0.00301% per day.
- The correlations of each asset are given below:

Correlation	ADS	VOW	Market Portfolio
ADS	1		
VOW	0.4227	1	
Market Portfolio	0.6689	0.8125	1

$W_{ADS} = 70\%$

$W_{VOW} = 30\%$

$\sigma_{ADS} = 1.794\%$

$\sigma_{VOW} = 1.251\%$

$r_m = 0.038\%$

$\sigma_m = 1.006\%$

$r_f = 0.00301\%$

$\rho_{ADS,VOW} = 0.4227$

$\rho_{ADS,m} = 0.6689$

$\rho_{VOW,m} = 0.8125$

Estimate the expected return of the shares in Adidas and Volkswagen as well as Hansel and Gretel's portfolio.

Question 2

Suppose there is a hedge fund VirtualUnited that had an average annual return of 9% p.a. and a standard deviation of 2% p.a. between 2015 – 2020. The beta of the hedge fund is 0.4. The average market return was 7% p.a. and the risk-free rate is 1% p.a.

a) Did the manager exhibit positive performance ability according to the CAPM? What is the manager's alpha?

b) What are the limitations of using alpha to measure performance?

Question 3

Your financial advisor has found four potential managed funds that fit your investment criteria. She has collected the following information about each one along with a relevant market index:

	Average return (% p.a.)	Standard Deviation (% p.a.)	Beta
S&P500 Market Index	10.7	19.0	1.00
Vanguard Value Fund	19.4	44.2	1.34
Invesco Small Cap Fund	13.5	21.8	1.12
iShares Growth Fund	12.9	17.6	1.30
Fidelity Large Cap Fund	9.3	5.6	0.71

Question 3

a) Calculate risk-adjusted measures of portfolio performance (Sharpe ratio, Treynor ratio and Jensen's alpha) for each of the funds and the market index if over the period the risk-free asset had a return of 5.6% p.a.

b) Which fund has exhibited the best performance over this period?

c) Which fund would be the best choice if:

(i) this would be the only risky asset held by you; or

(ii) this would be one of the many funds that you are combining into an actively managed portfolio?

Question 4

Steve N. Ross has been saving some extra money, which he would like to use for investing. He believes that the APT model is appropriate for pricing the following US stocks:

Factor	Expected Risk Premium (%)
Market	6.4%
Interest Rate	-0.6%
Yield Spread	5.1%

	Factor Risk Exposures		
	Market	Interest Rate	Yield Spread
Stock	β_1	β_2	β_3
Coca-Cola	1.0	-2.0	-0.2
IBM	1.2	0	0.3
Disney	0.3	0.5	1.0

Question 4

a) Calculate the expected return for the three stocks assuming a risk-free rate of 5%.

b) Steve N. Ross is considering creating a portfolio that has equal investments in all three stocks. What would be this portfolio's expected return?

c) Now he is thinking about investing $10,000 in Coca-Cola, $35,000 in IBM and $20,000 in Disney. What would be the sensitivity of this portfolio to each of the risk factors? What is the expected risk premium? What is the change in the portfolio's expected return?

Question 1

Hansel and Gretel escaped with all the witch's wealth and invested 70% of it in Adidas (ADS) and the remainder in Volkswagen (VOW) on the German Stock Exchange (Deutsche Borse Xetra).

- They used the daily returns of the most recent six months to calculate the risk (SD) of each share at 1.794% and 1.251%, respectively.
- From a relevant market portfolio, the expected daily return is 0.038% and the standard deviation of returns is 1.006%.
- The risk-free rate is 0.00301% per day.
- The correlations of each asset are given below:

Correlation	ADS	VOW	Market Portfolio
ADS	1		
VOW	0.4227	1	
Market Portfolio	0.6689	0.8125	1

$W_{ADS} = 70\%$

$W_{VOW} = 30\%$

$\sigma_{ADS} = 1.794\%$

$\sigma_{VOW} = 1.251\%$

$r_m = 0.038\%$

$\sigma_m = 1.006\%$

$r_f = 0.00301\%$

$\rho_{ADS,VOW} = 0.4227$

$\rho_{ADS,m} = 0.6689$

$\rho_{VOW,m} = 0.8125$

Estimate the expected return of the shares in Adidas and Volkswagen as well as Hansel and Gretel's portfolio.

Question 1 *continued …*

Estimate the expected return of the shares in Adidas and Volkswagen as well as Hansel and Gretel's portfolio.

Determine the stock betas using: $\beta_{i,m} = \dfrac{\sigma_{i,m}}{\sigma_m^2}$ where $\sigma_m = 1.006\%$

and $\sigma_{i,m} = \rho_{i,m}\,\sigma_i\,\sigma_m$

Adidas:

$$\sigma_{ADS,m} = \rho_{ADS,m}\sigma_{ADS}\sigma_m = 0.6689 \times 0.01794 \times 0.01006 = 0.0001207$$

$$\beta_{ADS,m} = \frac{0.0001207}{0.01006^2} = 1.19265$$

$$E(r_{ADS}) = r_f + \beta_{ADS}(ER_m - r_f)$$

$$= 0.0000301 + 1.19265 \times (0.00038 - 0.0000301)$$

$$= 0.0004474 = 0.04474\%\ per\ day$$

$W_{ADS} = 70\%$

$W_{VOW} = 30\%$

$\sigma_{ADS} = 1.794\%$

$\sigma_{VOW} = 1.251\%$

$r_m = 0.038\%$

$\sigma_m = 1.006\%$

$r_f = 0.00301\%$

$\rho_{ADS,VOW} = 0.4227$

$\rho_{ADS,m} = 0.6689$

$\rho_{VOW,m} = 0.8125$

Question 1 *continued …*

Volkswagon:

$$\sigma_{VOW,m} = \rho_{VOW,m}\sigma_{VOW}\sigma_m = 0.8125 \times 0.01251 \times 0.01006 = 0.0001023$$

$$\beta_{VOW,m} = \frac{0.0001023}{0.01006^2} = 1.01083$$

$$E(r_{VOW}) = r_f + \beta_{VOW}(ER_m - r_f)$$

$$= 0.0000301 + 1.01083 \times (0.00038 - 0.0000301)$$

$$= 0.0003838 = 0.03838\% \ per\ day$$

Portfolio:

$$ER_p = \sum_{i=1}^{n} w_i\mu_i \qquad = (0.7 \times 0.0004474) + (0.3 \times 0.0003838)$$

$$= 0.000428 = 0.0428\% \ per\ day$$

W_{ADS} = 70%
W_{VOW} = 30%
σ_{ADS} = 1.794%
σ_{VOW} = 1.251%
r_m = 0.038%
σ_m = 1.006%
r_f = 0.00301%
$\rho_{ADS,VOW}$ = 0.4227
$\rho_{ADS,m}$ = 0.6689
$\rho_{VOW,m}$ = 0.8125

Question 2

Suppose there is a hedge fund VirtualUnited that had an average annual return of 9% p.a. and a standard deviation of 2% p.a. between 2015 – 2020. The beta of the hedge fund is 0.4. The average market return was 7% p.a. and the risk-free rate is 1% p.a.

a) Did the manager exhibit positive performance ability according to the CAPM? What is the manager's alpha?

$$ER_{VU} = r_f + \beta_{VU}(ER_m - r_f) = 0.01 + 0.4 \times (0.07 - 0.01) = 0.034 = 3.4\%$$

$$Jensen's\ Alpha = ER_{VU} - [rf + \beta(ER_m - rf)]$$

$$= 0.09 - [0.01 + 0.4(0.07 - 0.01)] = 0.056 = 5.6\%$$

According to the CAPM, the expected return of VirtualUnited was 3.4% p.a. The average annual return is higher at 9% p.a., so the manager exhibited positive performance ability with an alpha of 5.6%.

Question 2 *continued …*

b) What are the limitations of using alpha to measure performance?

- Alpha does not tell us how much total risk is being taken, only beta risk even though a portfolio may not be fully diversified.
- Alpha is not forward looking, it can only be calculated once historical returns have been realised. If markets are totally efficient, and securities are properly priced at all times, it should not possible to identify and take advantage of mispricing and beat the market.
- Outperformance may also not be due to manager skill but market luck due to economic boom cycles.
- Past performance measured by alpha is not necessarily an indicator of future performance either.

Question 3

Your financial advisor has found four potential managed funds that fit your investment criteria. She has collected the following information about each one along with a relevant market index:

	Average return (% p.a.)	Standard Deviation (% p.a.)	Beta
S&P500 Market Index	10.7	19.0	1.00
Vanguard Value Fund	19.4	44.2	1.34
Invesco Small Cap Fund	13.5	21.8	1.12
iShares Growth Fund	12.9	17.6	1.30
Fidelity Large Cap Fund	9.3	5.6	0.71

Question 3 *continued* …

a) Calculate risk-adjusted measures of portfolio performance (Sharpe ratio, Treynor ratio and Jensen's alpha) for each of the funds and the market index if over the period the risk-free asset had a return of 5.6% p.a.

$$Sharpe\ ratio = \frac{ER_p - r_f}{\sigma_p} \qquad Treynor\ ratio = \frac{ER_p - r_f}{\beta_p} \qquad Jensen's\ alpha = ER_p - [rf + \beta(ER_m - rf)]$$

	Sharpe Ratio	Treynor Ratio	Alpha
S&P500 Market Index	$\frac{10.7 - 5.6}{19} = 0.268$	$\frac{10.7 - 5.6}{1} = 5.100$	$10.7 - [5.6 + 1(10.7 - 5.6)] = 0$
Vanguard Value Fund	$\frac{19.4 - 5.6}{44.2} = 0.312$	$\frac{19.4 - 5.6}{1.34} = 10.299$	$19.4 - [5.6 + 1.34(10.7 - 5.6)] = 6.966$
Invesco Small Cap Fund	$\frac{13.5 - 5.6}{21.8} = 0.362$	$\frac{13.5 - 5.6}{1.12} = 7.054$	$13.5 - [5.6 + 1.12(10.7 - 5.6)] = 2.188$
iShares Growth Fund	$\frac{12.9 - 5.6}{17.6} = 0.415$	$\frac{12.9 - 5.6}{1.30} = 5.615$	$12.9 - [5.6 + 1.30(10.7 - 5.6)] = 0.670$
Fidelity Large Cap Fund	$\frac{9.3 - 5.6}{5.6} = 0.661$	$\frac{9.3 - 5.6}{0.71} = 5.211$	$9.3 - [5.6 + 0.71(10.7 - 5.6)] = 0.079$

Question 3 *continued* …

b) Which fund has exhibited the best performance over this period?

The portfolio with the best performance will depend on what criteria you are measuring.

- The Vanguard Value Fund had the highest alpha indicating it had the highest realised return relative to the expected return from the CAPM, but likely occurred due to the higher total risk and standard deviation.
- The Vanguard Value Fund also has the highest Treynor ratio indicating it is most suitable to include as one of many funds to combine into an actively managed portfolio.
- However, the Fidelity Large Cap Fund has the highest Sharpe ratio, indicating it earned higher returns for each unit of risk taken.

Question 3 *continued* …

c) Which fund would be the best choice if:

(i) this would be the only risky asset held by you; or

(ii) this would be one of the many funds that you are combining into an actively managed portfolio?

- If this fund is the **only risky asset**, the Sharpe ratio is the preferred criterion for selecting the fund, and the Fidelity Large Cap Fund is preferred as it has the highest risk-adjusted returns.

- If the fund is **one of many combined into an actively managed portfolio**, the Treynor ratio is the preferred metric, and the Vanguard Value Fund is preferred as it earns the highest excess returns relative to beta.

Question 4

Steve N. Ross has been saving some extra money, which he would like to use for investing. He believes that the APT model is appropriate for pricing the following US stocks:

Factor	Expected Risk Premium (%)
Market	6.4%
Interest Rate	-0.6%
Yield Spread	5.1%

	Factor Risk Exposures		
	Market	Interest Rate	Yield Spread
Stock	β_1	β_2	β_3
Coca-Cola	1.0	-2.0	-0.2
IBM	1.2	0	0.3
Disney	0.3	0.5	1.0

Question 4 *continued* …

a) Calculate the expected return for the three stocks assuming a risk-free rate of 5%.

$$\boldsymbol{ER_i = r_f + \beta_1(r_1) + \beta_2(r_2) + \beta_3(r_3) = r_f + \beta_m(r_m) + \beta_{ir}(r_{ir}) + \beta_{ys}(r_{ys})}$$

$$\boldsymbol{ER_{KO}} = 0.05 + 1(0.064) + (-2)(-0.006) + (-0.2)(0.051) = 0.1158 = 11.58\%$$

$$\boldsymbol{ER_{IBM}} = 0.05 + 1.2(0.064) + (0)(-0.006) + 0.3(0.051) = 0.1421 = 14.21\%$$

$$\boldsymbol{ER_{DIS}} = 0.05 + 0.3(0.064) + 0.5(-0.006) + 1(0.051) = 0.1172 = 11.72\%$$

Question 4 *continued* …

b) Steve N. Ross is considering creating a portfolio that has equal investments in all three stocks. What would be this portfolio's expected return?

$$ER_P = \mu_P = w_1\mu_1 + w_2\mu_2 + w_3\mu_3$$

$$= \left(\frac{1}{3} \times 0.1158\right) + \left(\frac{1}{3} \times 0.1421\right)\left(\frac{1}{3} \times 0.1172\right) = 0.1250 = 12.50\%$$

Question 4 *continued* …

c) Now he is thinking about investing $10,000 in Coca-Cola, $35,000 in IBM and $20,000 in Disney. What would be the sensitivity of this portfolio to each of the risk factors? What is the expected risk premium? What is the change in the portfolio's expected return?

The sensitivity to each factor is dependent on the beta of the portfolio, which can be calculating by the betas for each stock based on their new portfolio weights:

$$\boldsymbol{Weight\ of\ Coca-Cola} = \frac{10{,}000}{10{,}000 + 35{,}000 + 20{,}000} = \frac{10{,}000}{65{,}000} = 0.154$$

$$\boldsymbol{Weight\ of\ IBM} = \frac{35{,}000}{65{,}000} = 0.538$$

$$\boldsymbol{Weight\ of\ Disney} = \frac{20{,}000}{65{,}000} = 0.308$$

Question 4 *continued* …

c) Now he is thinking about investing $10,000 in Coca-Cola, $35,000 in IBM and $20,000 in Disney. What would be the sensitivity of this portfolio to each of the risk factors? What is the expected risk premium? What is the change in the portfolio's expected return?

$$\boldsymbol{New\ Betas = w_1\beta_1 + w_2\beta_2 + \cdots + w_n\beta_n}$$

$$\boldsymbol{\beta_m} = 0.154(1) + 0.538(1.2) + 0.308(0.3) = 0.892$$

$$\boldsymbol{Expected\ RP_m} = 0.892 \times 0.064 = 0.057 = 5.7\%$$

$$\boldsymbol{\beta_{ir}} = 0.154(-2) + 0.538(0) + 0.308(0.5) = -0.154$$

$$\boldsymbol{Expected\ RP_{ir}} = -0.154 \times -0.006 = 0.0009 = 0.09\%$$

$$\boldsymbol{\beta_{ys}} = 0.154(-0.2) + 0.538(0.3) + 0.308(1) = 0.438$$

$$\boldsymbol{Expected\ RP_{ys}} = 0.438 \times 0.051 = 0.022 = 2.2\%$$

Question 4 *continued* …

c) Now he is thinking about investing $10,000 in Coca-Cola, $35,000 in IBM and $20,000 in Disney. What would be the sensitivity of this portfolio to each of the risk factors? What is the expected risk premium? What is the change in the portfolio's expected return?

$$ER_p = r_f + \beta_1(r_1) + \beta_2(r_2) + \beta_3(r_3) = r_f + RP_m + RP_{ir} + RP_{ys}$$

$$= 0.05 + 0.057 + 0.0009 + 0.022 = 0.1299 = 12.99\%$$

Portfolio return (equally weighted) = 12.50%

The change in portfolio expected return is **0.49%.**

The End

Next topic we look the Company Cost of Capital